The TCRen2 descriptor catalogue
hierarchy, classification, formulas and derivations

`tcren` 2.30.0 — 164 descriptors in 6 families

2 September 2026

Abstract

This appendix documents every descriptor `tcren` computes from a TCR–pMHC structure. It is organised around a claim: the 164 descriptors are not 164 ideas. They are a small set of *operators* applied to a smaller set of *objects*, each object a strict reduction of the atomic coordinates. Nine operators and eight objects account for the whole catalogue, and naming them makes three things visible that the family listing hides — which descriptors can be redundant with which (only within a reduction step), which are exactly determined by others, and which are invariant under what. Nineteen descriptors spread over four families turn out to be one formula, the Hill number, applied to seven different probability vectors at three orders. Section 5 treats MHC class I and class II, which condition several formulas and are unevenly represented in the corpus. Nothing here is fitted: every descriptor is a deterministic function of one structure.

Contents

1 Scope

A descriptor is a function $f : \mathcal{S} \to \mathbb{R}$ from one annotated TCR–pMHC structure to one number. Annotation means chain typing, CDR region markup and, for the MHC-facing columns, groove region markup; nothing else is required, and in particular no alignment to a reference and no prior canonicalisation of the input orientation. Each f carries a unit, an invariance class and, where something is known about its behaviour, a `STATUS` flag. Section 6 lists all 164 with their definitions; Section 7 lists the 27 flagged ones.

Three properties hold catalogue-wide and are assumed everywhere below.

No fitted parameter. No descriptor reads a label, a training set or a coefficient estimated from one. The statistical potentials are derived once from a structural database and shipped; the Potts couplings likewise. A descriptor's value on a structure does not depend on what else was in the batch.

Rigid-motion invariance. Every descriptor is invariant under the action of $SE(3)$ on the input coordinates. The frame-referred quantities of Section 3.8 achieve this by refitting their frame from the structure rather than inheriting one, so two copies of a complex in different laboratory orientations give identical rows.

Undefined is `NaN`, never zero. A descriptor with no support returns `NaN`. A contact order over one contacted residue, a standoff height over a face with no void, a stiffness over two springs: each is undefined, and 0 would rank as an extreme rather than as a missing value.

2 The hierarchy: a chain of reductions

The six *families* — placement, interface, topology, energetics, potts, kinetics — record which pass of which command produces a descriptor. That is a fact about the software. The fact about the science is which *object* the descriptor reads, and the objects form a chain in which each step discards information irreversibly (Figure 1).

$\mathcal{S}$, the coordinates. Around 10^4 heavy atoms with chain, residue and region markup. Everything derives from here; no descriptor reads it directly.

$\mathcal{F}$, the groove frame. An orthonormal triple $(\hat{u}, \hat{w}, \hat{n})$ obtained from the singular value decomposition of the MHC groove-floor Cα coordinates, with the signs fixed by the peptide's N→C direction and by which side the receptor sits on. $\mathcal{F}$ is refit from every structure, which is what makes the placement family comparable across inputs.

C, the contact set. Ordered pairs (i, j) of residues whose closest heavy atoms lie within 5 Å. This is the single free parameter almost everything downstream inherits.

$\mathcal{D}$, the distance maps. The Cα and Cβ matrices between two labelled regions. $\mathcal{D}$ keeps the metric and forgets the atoms.

$\mathcal{L}$, the labelled contacts. C with residue identity, or with a bond type (salt bridge, hydrogen bond, aromatic, hydrophobic) attached. $\mathcal{L}$ keeps chemistry and forgets geometry beyond the threshold.

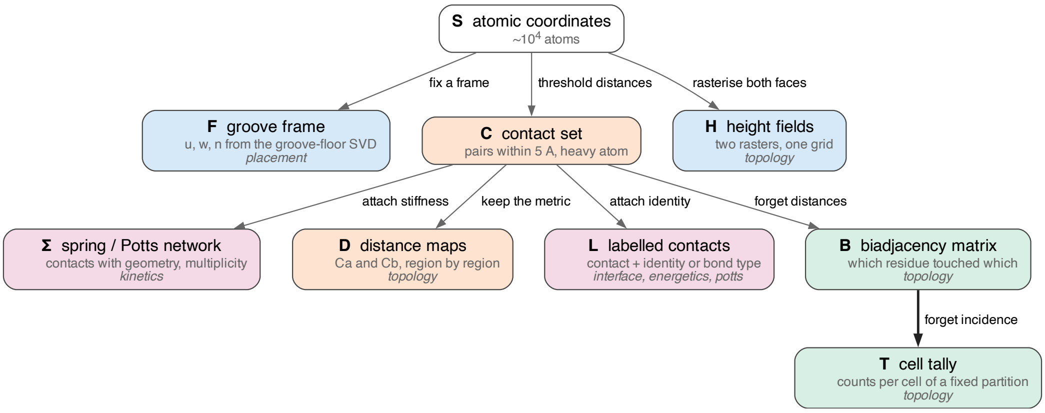

Figure 1: The reduction chain. Each arrow discards information; the bold arrow $\mathcal{B} \twoheadrightarrow \mathcal{T}$ is the one that matters most, because it is a quotient and therefore orders the redundancy structure of Section 2.1. The italic line in each node names the family that reads that object; *topology* appears in four, which is why it is the largest family and the least coherent one.

$\mathcal{B}$, the biadjacency matrix. $B \in \{0, 1\}^{m\times n}$ with the engaged CDR-loop residues as rows and the pMHC residues they touch as columns. $\mathcal{B}$ keeps *who touched whom* and forgets how far apart they were.

$\mathcal{T}$, the cell tally. Contact counts per cell of a fixed partition — the twelve cells of six CDR loops × {peptide, MHC}, or the twenty-four that split the peptide into three bands. $\mathcal{T}$ keeps how many and forgets which.

$\mathcal{H}$, the height fields. Two rasters $h_{\mathrm{pMHC}}(x, y)$ and $h_{\mathrm{TCR}}(x, y)$ on one shared grid in the $\mathcal{F}$ plane, with per-cell charge and hydropathy painted on.

Σ, the network. C with a stiffness or a Potts site attached to each pair: a spring network in the mechanical case, a set of occupiable sites in the Potts case.

2.1 The quotient, and what it predicts

$\mathcal{B} \twoheadrightarrow \mathcal{T}$ is a quotient map: the tally is a function of the biadjacency matrix, and many biadjacency matrices give the same tally. Every functional of $\mathcal{T}$ is therefore a functional of $\mathcal{B}$, and the converse fails. Two consequences follow without any measurement.

1. A $\mathcal{T}$-descriptor can be exactly determined by $\mathcal{B}$-descriptors, but no $\mathcal{B}$-descriptor can be exactly determined by $\mathcal{T}$-descriptors alone. Redundancy runs *down* the chain, never up it.
2. The parameter-free replacements are the ones further *up*. `g_comp_frac` counts components of the contact graph itself; `fp_b0_r7` counts components of a Cα flag complex at an arbitrary 7 Å. The first reads $\mathcal{B}$ and needs no radius; the second reads C again with a second threshold on top of the first. When both survive a redundancy screen they are measuring different things; when only one does, the chain says which to prefer.

Figure 2 draws the step for a small footprint.

3 The operators

Table 1 is the whole catalogue by what is applied rather than by what produced it. Nine operators suffice, and two of them — counts and frame coordinates — need no derivation, being the raw readings of $\mathcal{L}$ and

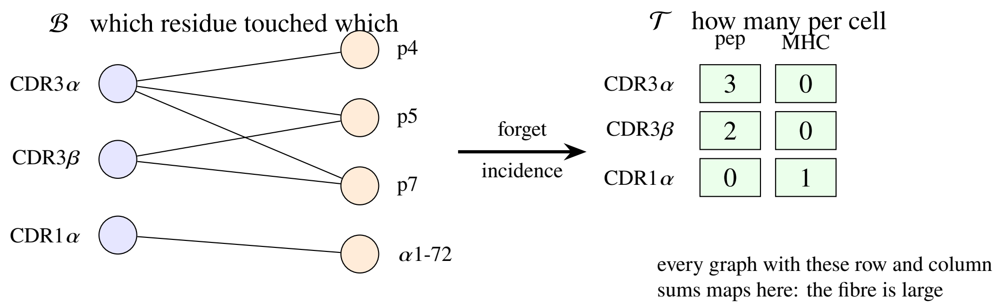

Figure 2: The quotient $\mathcal{B} \twoheadrightarrow \mathcal{T}$ on a six-contact footprint. The tally retains the row sums of the biadjacency matrix within each target block and discards which pMHC residue each contact reached, so `g_loop_overlap` — do the loops crowd onto the same residues — has no expression in $\mathcal{T}$ at all.

operator	object read	descriptors
count or share	C, $\mathcal{L}$	**49**
frame coordinate	$\mathcal{F}$	31
field moment	$\mathcal{H}$, $\mathcal{D}$	19
Hill number	$\mathcal{T}$, $\mathcal{B}$, spectra	19
potential sum	$\mathcal{L}$	19
homology invariant	$\mathcal{B}$, C	10
spectral invariant	$\mathcal{B}$, $\mathcal{D}$, Σ	9
correlation	$\mathcal{H}$, $\mathcal{D}$, $\mathcal{B}$	6
mechanical work	Σ	2
total		164

Table 1: The 164 descriptors by operator. Counts and shares dominate because the interface and kinetics families are largely tallies; the interesting structure is in the smaller rows.

$\mathcal{F}$ respectively. The remaining seven are derived below.

3.1 Hill numbers: 19 descriptors, one formula

This is the largest unification in the catalogue and the one worth stating first. Let p be a probability vector on k categories, $p_i \geq 0$, $\sum_i p_i = 1$. The *Hill number of order q* [5, 7] is the effective number of categories,

$$D_q(p) \;=\; \Big(\sum_{i\,:\,p_i>0} p_i^q\Big)^{1/(1-q)}, \qquad D_1(p) \;=\; \exp\Big(-\sum_i p_i \ln p_i\Big), \tag{1}$$

where D_1 is the $q \to 1$ limit. Three orders are used: D_0 is the richness, the count of occupied categories; D_1 is the exponentiated Shannon entropy; $D_2 = 1/\sum_i p_i^2$ is the reciprocal of Simpson's concentration, and discounts weakly populated categories. All three are in units of *categories*, which is what makes them comparable across partitions of different size, and each is bounded by k.

Four standard quantities are reparametrisations of Equation (1), not separate ideas:

$$\text{normalised Shannon entropy} \qquad H = \ln D_1 / \ln k, \tag{2}$$

$$\text{Pielou evenness} \qquad J = \ln D_1 / \ln D_0, \tag{3}$$

$$\text{participation ratio} \qquad \mathrm{PR} = D_2 / k, \tag{4}$$

$$\text{Gini–Simpson index} \qquad G = 1 - 1/D_2. \tag{5}$$

descriptor	p is over	q	normalisation
S_cell	the 12 cells, contact shares	0	none
D1_cell	the 12 cells	1	none
H_cell	the 12 cells	1	$\ln D_1/\ln 12$
D2_cell	the 12 cells	2	none
J_cell	the 12 cells	1	$\ln D_1/\ln D_0$
H_loop	the 6 CDR loops, contact shares	1	$\ln D_1/\ln 6$
D2_loop	the 6 CDR loops	2	none
D2_pep24	the 24 cells (peptide split in three bands)	2	none
pep_cov_even	L peptide positions, accessibility-discounted	1	$\ln D_1/\ln L$
pep_cov_d2n	the same	2	D_2/L
g_even_tcr	degrees of the m engaged TCR residues	1	$\ln D_1/\ln m$
g_even_pmhc	degrees of the n engaged pMHC residues	1	$\ln D_1/\ln n$
g_loop_even	distinct partners per CDR loop	1	$\ln D_1/\ln 6$
degree_evenness_tp	TCR-side degrees, TCR:peptide only	2	D_2/m
m_erank_tp	singular values of K_{tp}, normalised	2	$D_2/\min(L, M)$
m_erank_tm	singular values of K_{tm}, normalised	2	$D_2/\min(L, M)$
h0_pers_ent	minimum-spanning-tree edge lengths	1	$\ln D_1/\ln(n-1)$
partcoef_tcr	per-residue shares over {peptide, MHC}	2	$\text{mean}_i(1 - 1/D_2)$
partcoef_pmhc	per-residue shares over the 6 CDR loops	2	$\text{mean}_i(1 - 1/D_2)$

Table 2: 19 descriptors across four families as one formula: seven probability vectors, three orders, five normalisations. `S_cell` through `D2_pep24` read $\mathcal{T}$; `g_even_*` through `partcoef_*` read $\mathcal{B}$; the `m_erank_*` pair reads a spectrum of $\mathcal{D}$; `h0_pers_ent` reads a barcode of C.

Equation (3) explains the `STATUS` entry that `J_cell` is determined: it is $H_{\text{cell}} \ln 12/\ln S_{\text{cell}}$, a ratio of two shipped columns. Equation (4) and Equation (5) are the two that are *not* yet recorded in `STATUS` and follow from the algebra alone.

Table 2 instantiates the formula. Read a row as: take this probability vector, apply Equation (1) at this order, then apply this normalisation.

Three identities worth naming. Each follows from Equation (1) and can be checked in a line.

1. *Di Paola's participation coefficient is Gini–Simpson.* The published definition [2] is $P_i = 1 - \sum_s (k_{si}/k_i)^2$ over modules s, with k_{si} residue i's edge count into module s. Writing $p_s = k_{si}/k_i$ gives $\sum_s p_s^2 = 1/D_2(p)$, so $P_i = 1 - 1/D_2$ exactly — Equation (5). It is a diversity of order 2 on a per-residue module profile, which places it in Table 2 rather than beside it.

2. *Effective rank and cell entropy are the same operator.* `m_erank_tp` applies Equation (1) to the normalised singular-value spectrum of a proximity kernel; `D2_cell` applies it to a contact tally. They are not independent descriptor families and should not be presented as such — what differs is the input, not the measure.

3. *The shipped effective rank is order 2, not order 1.* Roy and Vetterli's Definition 1 [11] is $\text{erank}(A) = \exp\{H(p_1, \dots, p_Q)\}$ with $p_k = \sigma_k/\|\sigma\|_1$, $Q = \min\{M, N\}$ and H the Shannon entropy — that is exactly D_1 of the normalised spectrum. The shipped `m_erank_*` computes $(\sum_a s_a)^2/\sum_a s_a^2$, which is D_2 — the participation ratio, as the implementing function's own docstring says. Both are legitimate effective ranks and $D_2 \le D_1$ always; the name in the descriptor table points at the literature quantity while the code computes the order-2 sibling. Anyone comparing our column against a published effective rank must use $q = 2$.

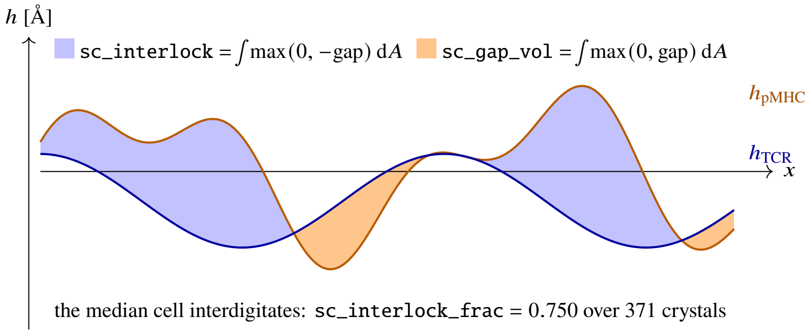

Figure 3: The gap field in cross-section. Blue is interdigitated volume, orange is void; the two are integrated separately because a mean over the retained cells cancels them against each other. On a real interface the blue area is the larger, by roughly 1,004 against 373 Å^3.

Why several orders ship at once. D_1 and D_2 rank differently. D_1 weights categories by their share; D_2 weights by the square, so a footprint with one dominant cell and a long tail of singletons scores much lower at $q = 2$ than at $q = 1$. Keeping both is the deliberate choice the library has made since the coverage block was written, and it is why `D2_pep24` rather than `H_cell` carries the shape channel.

3.2 Field moments, and the gap resolved by sign

The second operator takes moments of a scalar field defined on a shared index — grid cells for the surface block, contacting residue pairs for the map block. 19 descriptors, of which nine read one field: the gap between the two faces.

The two faces and their difference. $\mathcal{H}$ carries two height fields on *one* grid: $h_{\mathrm{pMHC}}(x, y)$, the highest pMHC surface point in cell (x, y), and $h_{\mathrm{TCR}}(x, y)$, the lowest receptor point in the same cell, both in the groove frame $(\hat{u}, \hat{w}, \hat{n})$ of Section 3.8 with $\hat{n}$ pointing at the receptor. Because they share a grid, the space between them is a subtraction rather than a new geometric construction:

$$\mathrm{gap}(x, y) \;=\; h_{\mathrm{TCR}}(x, y) - h_{\mathrm{pMHC}}(x, y) \quad [\text{Å}]. \tag{6}$$

A cell enters the comparison when both faces reach it and $|\mathrm{gap}| \le 10$ Å, cropped to a ± 12 Å window about the groove centre; the retained set is R with $|R| =$ `sc_cells` and area element $\delta A = 0.977$ Å^2.

The sign is not the one intuition suggests. A gap is naturally imagined as a void, so gap > 0. Measured over the crystal reference set it is predominantly negative: the receptor's lowest point in a cell lies *below* the groove's highest point in the same cell, because the two faces interdigitate rather than stack. Over 371 of the 374 crystals the mean interdigitated fraction is 0.750 (s.d. 0.096) and the mean gap -1.14 Å. Pooling the two signs into one mean therefore cancels the physics, which is why the field is reported three ways (Figure 3).

The nine gap descriptors are then

$$\text{pooled} \qquad \texttt{sc\_gap\_mean} = \frac{1}{|R|}\sum_R \text{gap}, \qquad \texttt{sc\_gap\_sd} = \text{sd}_R(\text{gap}), \tag{7}$$

$$\text{integrated} \qquad \texttt{sc\_gap\_vol} = \delta A \sum_R \max(0, \text{gap}), \qquad \texttt{sc\_interlock} = \delta A \sum_R \max(0, -\text{gap}), \tag{8}$$

$$\texttt{sc\_gap\_index} = \texttt{sc\_gap\_vol}/(|R|\,\delta A), \tag{9}$$

$$\text{by sign} \qquad \phi = \tfrac{1}{|R|}\big|\{\text{gap} < 0\}\big|, \qquad \texttt{sc\_gap\_depth} = -\overline{\text{gap}}\big|_{\text{gap}<0}, \tag{10}$$

$$\texttt{sc\_gap\_height} = \overline{\text{gap}}\big|_{\text{gap}>0}, \qquad \texttt{sc\_gap\_asym} = \frac{V_+ - V_-}{V_+ + V_-}, \tag{11}$$

with ϕ = `sc_interlock_frac`, V_+ = `sc_gap_vol` and V_- = `sc_interlock`. The decomposition is exact and is asserted by a unit test rather than assumed:

$$\texttt{sc\_gap\_mean} \;=\; (1-\phi)\cdot\texttt{sc\_gap\_height} \;-\; \phi\cdot\texttt{sc\_gap\_depth}. \tag{12}$$

Equation (12) is the reason the pooled mean is kept alongside the sign-resolved triple: it is their ϕ-weighted combination, so it is redundant with them by construction and is retained only as the familiar summary.

Attribution. `sc_gap_index` is the intensive form of the gap channel and takes the shape of Jones and Thornton's gap volume index [6], but ours divides by the retained contact area of the raster rather than by an interface accessible surface area, and the underlying field is a raster subtraction rather than a Voronoi gap volume. The channel is theirs; the quantity is not, and should not be cited as if it were.

The other field moments. `sc_dh`, `sc_dcharge` and `sc_dphobic` are mean absolute per-cell differences of the height, charge and Kyte–Doolittle fields; `sc_charge_prod` and `sc_phobic_prod` their mean per-cell products. `m_face_tp` and `m_face_tm` take a moment on a different index — contacting residue pairs rather than grid cells:

$$\texttt{m\_face} \;=\; \frac{1}{|C|}\sum_{(i,j)\in C}\big(\|\mathrm{C}\alpha_i - \mathrm{C}\alpha_j\| - \|\mathrm{C}\beta_i - \mathrm{C}\beta_j\|\big), \tag{13}$$

positive when the side chains lean towards each other and negative when the backbones are the close part and the side chains point away. Equation (13) is the one quantity in the catalogue that Cα coordinates alone cannot express, which is its reason for existing.

3.3 Correlation over a shared index

6 descriptors are a correlation coefficient between two vectors indexed by the same set. The operator is trivial; what varies is the index and the pair of fields, and the sign convention differs between them, which is the only thing a reader has to remember.

descriptor	index	fields	complementary when
`sc_shape`	retained grid cells	the two height fields	$r > 0$
`sc_charge`	retained grid cells	the two charge fields	$r < 0$
`sc_phobic`	retained grid cells	the two hydropathy fields	$r > 0$
`ca_cb_agreement_tp`	TCR:peptide shell pairs	Cα and Cβ maps	ρ high
`ca_cb_agreement_tm`	TCR:MHC shell pairs	Cα and Cβ maps	ρ high
`g_assort`	the E contacts	degrees at the two ends	—

`sc_charge` inverts because charge complementarity is plus meeting minus, whereas shape and apolar complementarity are like meeting like. `sc_shape` is Lawrence and Colman's shape complementarity idea

[8] carried onto a raster: their S_c is a median over dot-surface points of $(\mathbf{n}_A \cdot \mathbf{n}'_A) \exp(-wd^2)$ after trimming a peripheral band, and ours is a Pearson correlation of two height fields. The construction differs; the channel is the same one, and the resemblance should be stated that way rather than as a reimplementation.

`g_assort` is degree assortativity [10], correlating the degrees at the two ends of each edge. It is left undefined rather than zero when either side has constant degree: there is no correlation to measure, which is not the same as measuring no correlation. Assortativity is partly determined by the degree distribution itself [1], so it is not independent of `g_even_tcr` and `g_even_pmhc` and should be read beside them.

3.4 Spectral invariants

9 descriptors are eigenvalues or singular values of an operator built from geometry. Three constructions.

The normalised Laplacian. On the largest component of the bipartite contact graph, with A the adjacency matrix and D the degree matrix,

$$\mathcal{L} = I - D^{-1/2} A D^{-1/2}, \qquad \texttt{g\_alg\_conn} = \lambda_2(\mathcal{L}) \in [0, 2], \tag{14}$$

the algebraic connectivity [4]: near zero when the footprint is about to fall into two patches, so it is the continuous form of the Betti-0 count of Section 3.5. The normalisation matters. The combinatorial Laplacian's λ_2 scales with degree and would track interface size; Equation (14) is bounded. One caution follows from the same normalisation: $\lambda_{\max}(\mathcal{L}) = 2$ identically for *any* bipartite graph, and our contact graph is bipartite by construction, so $\lambda_{\max}$ carries no information here and must never be added as a descriptor.

The proximity kernel. For two labelled regions with Cα distance map $D \in \mathbb{R}^{L \times M}$, the Gaussian proximity kernel and its singular values $s_1 \geq s_2 \geq \cdots$ are

$$K = \exp\big(-D^{\odot 2}/2\sigma^2\big), \qquad \sigma = 5\text{ Å}, \qquad \texttt{m\_gap} = s_2/s_1. \tag{15}$$

σ is the contact criterion's own 5 Å, so the binary contact map is this kernel's $\sigma \to 0$ limit rather than an independent parameter. The entries of D cannot be compared across a 9-mer and a 15-mer, or across CDR3 loops of different length; its singular values can, which is the whole point of the construction. s_2/s_1 is near zero when the approach is separable — a rank-one K means a loop profile times a peptide profile, a receptor leaning on a surface rather than pairing specific residues with it. The corresponding D_2 of the same spectrum is `m_erank`, Section 3.1.

The stiffness tensor. Each inter-body residue contact is a Hookean spring anchored at the two Cα atoms, with stiffness k_i from heavy-atom-pair multiplicity divided by squared distance and unit direction $\hat{u}_i$. The interface linear-response stiffness is

$$K = \sum_i k_i\, \hat{u}_i \otimes \hat{u}_i \in \mathbb{R}^{3\times 3}, \qquad \texttt{S\_tot} = \operatorname{tr} K, \quad \texttt{lam\_max}, \texttt{lam\_min} = \lambda_{\max,\min}(K). \tag{16}$$

`K_tens` is the component along the docking axis, `K_shear` = tr K − `K_tens` the in-plane remainder and `aniso` their ratio; the last three are exactly determined by the first two and are flagged as such. No potential enters Equation (16) — it is geometry and multiplicity only, which is what makes `rupture_work` an off-rate proxy independent of the energy channel it is compared against.

3.5 Homology invariants

10 descriptors count connected components and independent cycles, on two different complexes, and the difference between the two complexes is the whole story.

The Cα flag complex. Take the contacted pMHC Cα atoms, join any pair within r, and build the flag (clique) complex. `fp_b0_r7` and `fp_b1_r7` are its Betti numbers at $r = 7$ Å, `fp_chi_r7` $= b_0 - b_1$ the Euler characteristic — determined, and flagged — and `fp_b0_frac_r7` is b_0 per contacted residue, the size-free form. The same four at $r = 8$ Å.

The contact graph itself. `g_comp_frac` $= C/V$ and

$$\texttt{g\_cyclo\_frac} \;=\; \frac{E - V + C}{E}, \tag{17}$$

with E contacts, $V = m + n$ residues and C components. The numerator of Equation (17) is the cyclomatic number, which for a graph is the first Betti number; dividing by E is what makes it size-free, and the footprint module rejects the raw form precisely because with of order thirty contacts among of order thirty residues it is dominated by E and simply tracks interface size.

Which to prefer. The flag-complex pair introduces a *second* threshold on top of the 5 Å contact criterion, and 7 and 8 Å are not derived from anything. The contact graph pair introduces none. By the ordering of Section 2.1 the parameter-free form is the one to prefer where both are informative — which is a statement about parsimony, not about measured performance; the flag-complex columns carry benchmark numbers and are not being retired here.

Persistence entropy. `h0_pers_ent` belongs to this section by provenance and to Section 3.1 by formula. The H_0 barcode's death times are exactly the minimum spanning tree's edge lengths, so no filtration library is needed: build the tree, normalise the edge lengths to a probability vector, take Equation (2). It is scale-free by construction.

3.6 Potential sums

19 descriptors sum a pair potential over a contact set, or take a difference of two such sums. The Hamiltonian is

$$\Phi \;=\; c_{\mathrm{TP}}\,\Phi_{\mathrm{TCR:pep}} \;+\; c_{\mathrm{TM}}\,\Phi_{\mathrm{TCR:MHC}} \;+\; c_{\mathrm{PM}}\,\Phi_{\mathrm{pep:MHC}}, \qquad \Phi_I \;=\; \sum_{(i,j)\in C_I} e_I(a_i, a_j), \tag{18}$$

where C_I is interface I's contact set, a_i the amino acid at residue i, and e_I that interface's potential — TCRen2 for TCR:peptide, Miyazawa–Jernigan [9] for the two presentation interfaces. The coefficients exist because the three interfaces are scored with *different potentials on different scales*; each term enters divided by that interface energy's spread over the crystal reference set, $c_I = 1/\mathrm{sd}(\Phi_I)$. Lower Φ is more favourable throughout.

What cancels. Because no interface energy carries a within-chain term, varying the peptide leaves $\Phi_{\mathrm{TCR:MHC}}$ untouched and varying the TCR leaves $\Phi_{\mathrm{pep:MHC}}$ untouched — exactly, not approximately, since a truncated side chain moves no atom of the other two chains. This is why a peptide-ranking task reads $\Delta\Phi_{\mathrm{TCR:pep}}$ and a receptor-ranking task does not read $\Delta\Phi_{\mathrm{pep:MHC}}$ at all.

The reference difference. $\Delta\Phi$ (written `dPhi_*`) is the difference of Φ against a reference *sequence*, never a derivative: $\Delta\Phi = \Phi(\text{sequence}) - \Phi(\text{reference})$. In the hard form the reference is poly-alanine, one arbitrary sequence. In the smoothed form it is the free energy of the residue *background*, which is a distribution rather than a choice. Since Φ is a sum of independent local fields over the varying positions,

$$\varphi_i(a) \;=\; \sum_I c_I \sum_{j\,:\,(i,j)\in C_I} e_I(a, y_j), \tag{19}$$

with y the frozen partners, the partition function factorises exactly and the reference free energy is available in closed form:

$$\Phi_{\mathrm{ref}} \;=\; -\frac{1}{\beta}\sum_i \log \sum_a p(a)\, e^{-\beta\varphi_i(a)}, \qquad \Delta\Phi \;=\; \Phi(\text{observed}) - \Phi_{\mathrm{ref}}, \tag{20}$$

p being the background composition over the twenty amino acids. The second cumulant of the same log partition function is the variance descriptor,

$$\operatorname{Var}\Phi \;=\; \sum_i \operatorname{Var}_\beta[\varphi_i], \qquad \operatorname{Var}_\beta \;\text{under}\; \frac{p(a)e^{-\beta\varphi_i(a)}}{\sum_b p(b)e^{-\beta\varphi_i(b)}}. \tag{21}$$

Equation (21) asks how sharply a position's energy responds to residue identity at all: a position whose twenty fields are equal contributes to neither $\Delta\Phi$ nor $\operatorname{Var}\Phi$; one with a single strongly preferred residue contributes to both. It is a second cumulant, not a difference of differences, and the catalogue carries no dd quantity — the only one in the package is $\Delta\Delta G$, the change in binding free energy on mutation, which is not a descriptor.

3.7 The Potts family

Five descriptors read the same interface against a partition function rather than against a reference sequence. The configuration is the contact map itself: sites a are the residue pairs that *could* have contacted (Cα within a radius) and $\sigma_a = 1$ iff a heavy-atom contact formed. With one-body fields η and couplings A between neighbouring cells of the map,

$$E(\sigma) = -\sum_a \eta_a\sigma_a - \tfrac{1}{2}\sum_{a,b} A_{ab}\sigma_a\sigma_b, \qquad P(\sigma) = e^{-E(\sigma)}/Z. \tag{22}$$

`neg_energy` is $-E(\sigma^{\mathrm{obs}})$, `log_z` is $\log Z$ over every contact map the geometry admits — the interface's *capacity* — and `log_lik` is $\log P(\sigma^{\mathrm{obs}})$, its *typicality*. `psi` is `log_lik` per available site, so interfaces of different size compare. The three are not independent:

$$\texttt{neg\_energy} \;=\; \texttt{log\_z} + \texttt{log\_lik}, \tag{23}$$

directly from Equation (22), and Equation (23) is the `STATUS` entry on `neg_energy`. Fitting is by penalised pseudolikelihood and needs no Z; scoring obtains Z by annealed importance sampling from the uncoupled model, whose partition function is exact.

3.8 Frame coordinates

The 31 placement descriptors are the rigid-body pose and the two CDR3 loops expressed in the groove frame. They are the only family that needs a frame at all, and the frame is refit from every structure (Figure 4).

Six of them are the rigid-body pose: `dock_d` the stub-to-stub separation, `dock_torsion` the dihedral of the TCR about the MHC stub, and the four direction cosines `dock_tcr_uy`, `dock_tcr_uz`, `dock_mhc_uy`, `dock_mhc_uz`. Four more locate the CDR centroid: `height` above the groove plane, `shift_u` and `shift_w` in plane, and $\texttt{offset} = \sqrt{\texttt{shift\_u}^2 + \texttt{shift\_w}^2}$, which is determined by the two and flagged. Twenty are the two CDR3 loops, each contributing a reach, three position cosines, three axis cosines, an engagement depth and an extension.

The crossing angle, and one banned column. `crossing_signed` is the angle between the V$\alpha \to$ Vβ axis projected into the groove plane and the groove long axis, on $[-180°, 180°)$; its *sign* is the docking polarity, canonical or reversed, and $\texttt{crossing} = |\texttt{crossing\_signed}|$ is determined by it. Polarity and

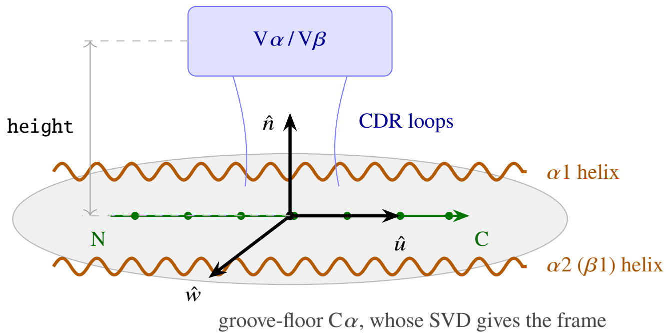

Figure 4: The groove frame. $\hat{u}$, $\hat{w}$ and $\hat{n}$ come from the singular value decomposition of the groove-floor Cα coordinates, with signs fixed by the peptide's N→C direction and by which side the receptor sits on. Every placement descriptor is a length, an angle or a direction cosine in this frame, so all thirty-one are $SE(3)$-invariant despite reading a frame.

angle are different quantities and conflating them is a category error: docking polarity is stereotyped across complexes whose crossing angles vary widely [3], and reversed-polarity receptors bind but cannot signal [12] — a functional discriminator no energy term sees.

`pitch` is the incident angle out of the groove plane. It is computed, catalogued and **banned as a feature**: it reproduces no clean geometric angle yet out-discriminates every one of them, which is contamination by the generator's confidence rather than interface geometry. The library raises if a scoring channel names it.

4 Classification: three axes and an invariance signature

The catalogue ships one classification — the five invariance classes, 65 geometric, 64 compositional, 20 energetic, 14 topological and 1 categorical. It is useful and it is coarse. Two descriptors computed by the *same operator* can land in different classes because the underlying field differs: `sc_shape` is a Pearson correlation of two height fields and is called geometric, while `sc_charge` is a Pearson correlation of two charge fields and is called compositional. Both readings are defensible; neither is derivable from the formula alone.

A sharper classification is a *triple*, and it is read off the definition rather than asserted:

$$f \longmapsto \big(\text{object} \in \textit{Section } 2,\quad \text{operator} \in \textit{Section } 3,\quad \text{invariance signature}\big). \tag{24}$$

The signature is a four-bit vector recording which transformations leave f unchanged. Only the first is a group action in the usual sense; the others are the practical questions a modeller asks.

bit	holds when	what it decides
SEQ	substituting residue identities at fixed coordinates leaves f unchanged	separates shape from chemistry. Every `Phi_*`, `sc_charge`, `sc_phobic` and `ct_*` column fails it; every placement, homology and gap column passes
SIZE	f is unchanged by scaling the interface up at fixed shape	intensive against extensive. This is the axis the size-confound screen measures, and the one a normalisation is meant to fix
ORD	permuting positions within a chain leaves f unchanged	fails for `co_pep`, `co_mhc`, `pep_cov_centre` and `pep_cov_spread`, which read sequence position; holds for every pure tally
MET	f depends on the contact set only, not on distances	holds for the homology and Hill blocks read off $\mathcal{B}$ or $\mathcal{T}$, fails for every moment of $\mathcal{D}$ or $\mathcal{H}$

Two uses. First, the signature predicts redundancy where the shipped class does not: two descriptors with the same object, same operator and same signature are near-certain duplicates, and the catalogue contains several such pairs by design — `H_cell` and `D2_cell` differ only in q. Second, it makes the *intended* contrasts visible. `m_erank_tp` and `m_gap_tm` share an object and an operator but differ sharply in how much peptide length they carry, so a model given both can form the contrast that cancels the shared part. Keeping a length-coupled column beside a length-free one is deliberate; the `STATUS` entries of Section 7 say which is which so the choice is the caller's.

What the shipped classes are, then. The five-class `INVARIANCE` labelling is the projection of the triple onto its most useful single coordinate, and is kept because a one-word label is what a table column can hold. Section 6 prints it beside the operator so both readings are available on the same row.

5 MHC class I and class II

Several formulas above are conditioned on which class of MHC presents the peptide, and one descriptor — `mhc_class_bin`, the single categorical column — exists to carry that condition. This section states what differs, what it does to the descriptors, and what the corpus actually holds.

5.1 What differs structurally

The class I groove is closed at both ends: the peptide's N- and C-termini are pinned in the A and F pockets, so a peptide longer than the canonical 8–10 residues cannot extend and must *bulge* out of the groove. The class II groove is open at both ends: the peptide runs through in an extended polyproline-II-like conformation with overhangs at either end, register set by anchor pockets rather than by termini, and length is not constrained the same way. Measured on our own crystal reference set, class I peptides have median length 9 (interquartile range 9–10, full range 4–13) and class II median 13 (interquartile range 12–15, full range 6–20).

The region vocabulary differs too, and this catches callers out. Class I presents from a single heavy chain whose $\alpha 1$ and $\alpha 2$ domains supply both groove helices (`HELIX_A1`, `HELIX_A2`). Class II presents from *two* chains, $\alpha 1$ from the α chain and $\beta 1$ from the β chain (`HELIX_A1`, `HELIX_B1`). That is why the helix vocabulary has three entries rather than two, and why the MHC-facing descriptors are computed over a three-region set of which one is always absent.

5.2 What it does to the descriptors

The 24-cell partition is class I–shaped. `D2_pep24` splits the peptide at fixed positions 3 and 6, which are thirds of a 9-mer and are not thirds of a 15-mer. The descriptor is well defined on class II and means something different there.

The length-normalised block is what puts the two classes on one scale. Every `pep_cov_*` column divides by the peptide's own length L, which is the reason that block exists: a class I 8-mer, a class I 13-mer and a class II 18-mer become comparable.

The raster is sized for class II. The surface grid's default extent is sized for a class II 15-mer with overhangs, which is much wider than any receptor footprint; the comparison window is cropped to ± 12 Å for exactly that reason.

The bulge mechanism is class I–specific. `m_erank_tp` is peptide-length coupled (Spearman -0.547 on 148 class I crystals) because in a closed groove a longer peptide must bulge and a bulged peptide presents more independent approach modes. That mechanism does not apply in an open groove, so the same correlation in class II would have a different cause.

A coefficient fitted across both classes without the indicator is fitted to a mixture. This is the reason `mhc_class_bin` is catalogued at all. It is one of the 5 columns computed without the receptor, so it conditions the other descriptors rather than being scored beside them.

descriptor	class I	class II	reading
peptide length L (median)	**9**	**13**	closed groove against open
n_pep_contacted	7	8	the receptor reads a near-constant window
pep_cov_frac	0.700	0.600	so the *share* falls as L rises
pep_cov_spread	0.330	0.423	contacts spread further along a longer peptide
co_pep	0.259	0.219	contact order falls with the longer span
burial (Å^2)	1842	1960	a larger interface
sc_gap_mean (Å)	−1.217	−1.313	interdigitation holds in both
sc_interlock_frac	0.769	0.762	and at the same magnitude
sc_coverage	0.953	0.937	the raster covers both
m_erank_tp	0.328	0.305	
g_loop_overlap	0.098	0.085	

Table 3: Medians by MHC class over the 374 crystals of the reference set: 280 class I and 94 class II. The first block is where the classes separate and the second is where they do not — the gap channel of Section 3.2 reads the same on both, so the interdigitation result is not a class I artefact.

5.3 What the corpus holds, and the honest limit

Table 3 is the measured contrast over the 374-crystal reference set, the only part of the corpus that contains class II at all.

Two readings are worth drawing out. n_pep_contacted moves from 7 to 8 while L moves from 9 to 13: the receptor reads a roughly constant-size window of peptide whatever the groove class, so an un-normalised peptide count reads groove class as much as it reads receptor behaviour, and the length-normalised block is not a convenience but a correction. And the gap channel is class-invariant to two decimal places, which means the interdigitation finding of Section 3.2 is a property of TCR–pMHC interfaces rather than of class I ones.

The limit, stated once. Class II appears in the corpus *only* among the crystals — 94 of 374 structures. Every modelled set is entirely class I, so mhc_class_bin is constant on all of them, and it is constant on both receptor-ranking benchmarks for the same reason. Any class II claim therefore rests on 94 structures and cannot be checked against a modelled set at all.

6 The catalogue

Generated from tcren.recognition at version 2.30.0. Each row carries the shipped invariance class, the operator of Section 3, the units and the definition; a dagger marks a STATUS flag.

6.1 The placement family (31 descriptors)

The rigid-body pose and the two CDR3 loops, read in the groove frame.

descriptor	class	operator	units	definition
pitch†	geometric	frame coordinate	°	Incident angle of the TCR out of the groove plane. **Banned as a feature**: it reproduces no clean geometric angle yet out-discriminates every one of them, which is AlphaFold-confidence contamination rather than geometry.
crossing†	geometric	frame coordinate	°	Crossing (scanning) angle between the Vα->Vβ axis projected into the groove plane and the groove long axis.

descriptor	class	operator	units	definition
crossing_signed	geometric	frame coordinate	°	The same angle on [-180, 180); its sign is the docking polarity, canonical or reversed.
dock_d	geometric	frame coordinate	Å	MHC-stub to TCR-stub rigid-body separation.
dock_torsion	geometric	frame coordinate	rad	Rigid-body dihedral of the TCR about the MHC stub; the docking twist. Circular, wraps at +-pi.
dock_tcr_uy	geometric	frame coordinate	cosine	y component of the TCR stub unit vector in the MHC frame.
dock_tcr_uz	geometric	frame coordinate	cosine	z component of the TCR stub unit vector; how high the receptor body rides over the groove.
dock_mhc_uy	geometric	frame coordinate	cosine	y component of the MHC stub unit vector.
dock_mhc_uz	geometric	frame coordinate	cosine	z component of the MHC stub unit vector.
height	geometric	frame coordinate	Å	Elevation of the CDR Cα centroid above the groove plane.
shift_u	geometric	frame coordinate	Å	In-plane displacement of that centroid from the peptide centroid along the groove long axis.
shift_w	geometric	frame coordinate	Å	The same along the groove short axis.
offset[†]	geometric	frame coordinate	Å	Length of the in-plane displacement; lateral shift whatever its direction.
cdr3a_reach	geometric	frame coordinate	Å	Distance from the loop's Cα centroid to the peptide Cα centroid; how far CDR3α reaches.
cdr3a_ou	geometric	frame coordinate	cosine	Where CDR3α sits over the groove, along the long axis u.
cdr3a_ow	geometric	frame coordinate	cosine	Where CDR3α sits over the groove, along the short axis w.
cdr3a_on	geometric	frame coordinate	cosine	Where CDR3α sits over the groove, along the groove normal n.
cdr3a_au	geometric	frame coordinate	cosine	Orientation of CDR3α's N->C axis against the groove long axis u.
cdr3a_aw	geometric	frame coordinate	cosine	Orientation of CDR3α's N->C axis against the groove short axis w.
cdr3a_an	geometric	frame coordinate	cosine	Orientation of CDR3α's N->C axis against the groove normal n.
cdr3a_topep	geometric	frame coordinate	Å	Minimum Cα-Cα distance from CDR3α to the peptide; its engagement depth.
cdr3a_ext	geometric	frame coordinate	Å	End-to-end extension of CDR3α, the Cα_N to Cα_C distance.
cdr3b_reach	geometric	frame coordinate	Å	Distance from the loop's Cα centroid to the peptide Cα centroid; how far CDR3β reaches.
cdr3b_ou	geometric	frame coordinate	cosine	Where CDR3β sits over the groove, along the long axis u.
cdr3b_ow	geometric	frame coordinate	cosine	Where CDR3β sits over the groove, along the short axis w.
cdr3b_on	geometric	frame coordinate	cosine	Where CDR3β sits over the groove, along the groove normal n.
cdr3b_au	geometric	frame coordinate	cosine	Orientation of CDR3β's N->C axis against the groove long axis u.

descriptor	class	operator	units	definition
cdr3b_aw	geometric	frame coordinate	cosine	Orientation of CDR3β's N->C axis against the groove short axis w.
cdr3b_an	geometric	frame coordinate	cosine	Orientation of CDR3β's N->C axis against the groove normal n.
cdr3b_topep	geometric	frame coordinate	Å	Minimum Cα-Cα distance from CDR3β to the peptide; its engagement depth.
cdr3b_ext	geometric	frame coordinate	Å	End-to-end extension of CDR3β, the Cα_N to Cα_C distance.

6.2 The interface family (26 descriptors)

How much contact there is, and of what chemical kind.

descriptor	class	operator	units	definition
burial	geometric	count or share	Å^2	Interface buried surface, SASA(TCR) + SASA(pMHC) - SASA(complex), by Shrake-Rupley.
extent	compositional	count or share	count	Distinct TCR residues contacting the pMHC over both receptor interfaces.
chain_balance	compositional	count or share	fraction	min(a,b)/(a+b) over TCR:peptide contacts by chain; 0.5 when both chains engage equally, 0 when only one does.
n_contacts_tp	compositional	count or share	count	TCR-peptide residue-residue contacts.
n_contacts_tm	compositional	count or share	count	TCR-MHC residue-residue contacts.
n_pep_contacted	compositional	count or share	count	Distinct peptide residues the TCR contacts.
n_hbond	compositional	count or share	count	Polar N/O atom pairs within 3.5 A across TCR:peptide.
ct_tp_salt_bridge†	compositional	count or share	count	Cationic-N / anionic-O pairs within 4 A across TCR:peptide.
ct_tp_aromatic	compositional	count or share	count	Ring-atom pairs between aromatic residues across TCR:peptide.
ct_tp_hydrophobic	compositional	count or share	count	Apolar C-C pairs between apolar residues across TCR:peptide.
ct_tp_other	compositional	count or share	count	Remaining classified TCR:peptide contacts.
ct_tm_salt_bridge	compositional	count or share	count	Salt bridges across TCR:MHC.
ct_tm_hydrogen_bond	compositional	count or share	count	Hydrogen bonds across TCR:MHC.
ct_tm_aromatic	compositional	count or share	count	Aromatic contacts across TCR:MHC.
ct_tm_hydrophobic	compositional	count or share	count	Hydrophobic contacts across TCR:MHC.
ct_tm_other	compositional	count or share	count	Remaining classified TCR:MHC contacts.
cdr3_dominance	compositional	count or share	fraction	CDR3(α+β) share of all CDR TCR:peptide contacts.
cdr3_ab_imbalance	compositional	count or share	fraction	abs(CDR3a - CDR3b) / (CDR3a + CDR3b); how one-sided the CDR3 engagement is.
chain_cdr_imbalance	compositional	count or share	fraction	abs(a - b) / (a + b) over all CDR contacts; the chain-level mirror of chain_balance.
n_clashes	compositional	count or share	count	Peptide-partner heavy-atom pairs overlapping by more than 0.4 A on Bondi radii.

descriptor	class	operator	units	definition
clash_score	geometric	field moment	Å	Summed overlap depth of those clashing pairs; the steric burden of a forced pose.
mhc_class_bin†	categorical	count or share	class I / II	Which class of MHC presents the peptide: 0 for class I, 1 for class II. Class I and class II grooves differ in shape and in how they hold a peptide, so this conditions the other descriptors rather than being scored beside them; a coefficient fitted across both classes without it is fitted to a mixture.
n_loop_contacts†	compositional	count or share	count	Contacts the six-CDR-loop partition sees; framework contacts are outside it by construction.
n_pep_contacts	compositional	count or share	count	Of those loop contacts, the ones reaχng the peptide.
n_mhc_contacts	compositional	count or share	count	Of those loop contacts, the ones reaχng the MHC.
n_pep_int†	compositional	count or share	count	Intra-peptide residue contacts, at 5 A with a sequence separation of at least three.

6.3 The topology family (70 descriptors)

The footprint: the contact graph, the cell tally, the two height fields, and the length-agnostic invariants of the Cα/Cβ maps.

descriptor	class	operator	units	definition
H_cell	compositional	Hill number	fraction	Normalized Shannon entropy of the contact composition over the twelve cells (six CDR loops x {peptide, MHC}).
D1_cell†	compositional	Hill number	count	Hill number of order 1 over the same cells, exp(H); monotone in H_cell.
D2_cell	compositional	Hill number	count	Hill number of order 2, 1/sum(p^2); the effective number of engaged cells, discounting the weakly populated ones.
S_cell	compositional	Hill number	count	Richness: how many of the twelve cells are occupied.
J_cell†	compositional	Hill number	fraction	Pielou evenness over the occupied cells. NaN when one cell is occupied.
H_loop	compositional	Hill number	fraction	Normalized entropy over the six CDR loops alone, ignoring which target each contact reaches.
D2_loop	compositional	Hill number	count	Hill number of order 2 over the six loops.
D2_pep24	compositional	Hill number	count	Hill number of order 2 over the twenty-four-cell partition, the peptide split into N-terminal, central and C-terminal bands.
ab_imb	compositional	count or share	signed fraction	Signed (TRA - TRB)/(TRA + TRB) over CDR-loop contacts; positive is α-shifted.
ab_imb_pep	compositional	count or share	signed fraction	The same restricted to peptide-side contacts.
ab_imb_mhc	compositional	count or share	signed fraction	The same restricted to MHC-side contacts.

descriptor	class	operator	units	definition
L_canon	compositional	count or share	log-odds	Canonical-docking log odds-ratio of loop class (germline, CDR3) against target (MHC, peptide), Haldane-Anscombe corrected. High when CDR3 sits on the peptide and the germline loops on the helices.
p_germ_mhc	compositional	count or share	fraction	Share of germline (CDR1/CDR2) contacts that reach the MHC.
p_cdr3_pep	compositional	count or share	fraction	Share of CDR3 contacts that reach the peptide.
pep_free_frac	compositional	count or share	fraction	Share of the peptide the groove leaves for the receptor: mean over positions of n_TCR/(n_TCR + n_MHC). The threshold-free reading of 'peptide without its MHC anchors'.
pep_cov_frac	compositional	count or share	fraction	Peptide positions the TCR contacts, over peptide length.
pep_cov_even	compositional	Hill number	fraction	Pielou evenness of the accessibility-discounted contact distribution, base ln(peptide length); how evenly the receptor uses the peptide it can reach.
pep_cov_d2n	compositional	Hill number	fraction	Hill number of order 2 of that distribution over peptide length; the effective share of the peptide engaged.
pep_cov_centre	compositional	count or share	fraction	Contact-weighted mean position on [0, 1] from N- to C-terminus; 0.5 is centred.
pep_cov_spread	compositional	count or share	fraction	Contact-weighted standard deviation of that position, doubled; approaches 1 when the receptor reaches both termini.
h0_pers_ent	geometric	Hill number	fraction	Normalized entropy of the H0 barcode of the contacted pMHC Cα cloud. The bar lengths are the minimum spanning tree's edges, so no filtration is chosen.
g_even_tcr†	topological	Hill number	fraction	Pielou evenness of the contact degrees of the engaged CDR-loop residues, base the engaged count. 1 when every engaged residue carries the same number of partners.
g_even_pmhc	topological	Hill number	fraction	Pielou evenness of the contact degrees of the engaged pMHC residues, base the engaged count.
g_comp_frac†	topological	homology invariant	fraction	Connected components of the bipartite contact graph per node; the parameter-free form of the footprint patch count, needing no Cα radius.
g_alg_conn	topological	spectral invariant	ratio	Second-smallest eigenvalue of the normalised Laplacian on the largest contact-graph component, in [0, 2]. Near 0 when the footprint is about to fall into two patches.
g_cyclo_frac	topological	homology invariant	fraction	Contacts beyond a spanning forest over all contacts, (E - V + C) / E, which is the contact graph's first Betti number made size-free. High when the footprint is interlocked.

descriptor	class	operator	units	definition
g_loop_even†	compositional	Hill number	fraction	Pielou evenness over the six CDR loops of the number of distinct pMHC residues each reaches, base 6. Counts partners rather than contacts, so it does not track residue size.
g_loop_overlap	compositional	count or share	fraction	Mean pairwise Jaccard overlap of the engaged CDR loops' pMHC partner sets. High when the loops crowd onto the same residues instead of partitioning the surface.
g_assort	topological	correlation	ratio	Degree assortativity of the contact graph: the correlation, over contacts, between the degrees of the two residues involved.
degree_evenness_tp†	compositional	Hill number	fraction	Participation ratio of the receptor-side contact degrees across TCR:peptide, in [0, 1]. Low when a few over-reaχng side chains hoard the contact budget.
frac_well_coordinated_tp†	compositional	count or share	fraction	Share of contacting receptor residues reaχng no more than three peptide residues, the count a crystal side chain typically makes.
m_erank_tp†	geometric	Hill number	fraction	Effective rank of the CDR-loop x peptide Cα proximity kernel over its maximum: how many independent approach modes the interface has. Peptide-length coupled (Spearman -0.547 on 148 class I crystals) because a longer class I peptide bulges.
m_gap_tp†	geometric	spectral invariant	ratio	Second over first singular value of that kernel. Near 0 when the approach is separable into a loop profile times a peptide profile rather than pairing specific residues.
m_erank_tm†	geometric	Hill number	fraction	Effective rank fraction of the CDR-loop x MHC-helix kernel. CDR3-length coupled (Spearman -0.437), so it reads how much loop there is to spread over the helices.
m_gap_tm	geometric	spectral invariant	ratio	Second over first singular value of the CDR-loop x MHC-helix kernel. The least length-coupled column in the catalogue: -0.012 against CDR3 length, +0.027 against peptide length, 99.9 per cent of its variance surviving both.
m_face_tp	geometric	field moment	Å	Mean Cα-Cα minus Cβ-Cβ distance over the contacting TCR:peptide residue pairs. Positive when side chains lean towards each other, negative when the backbones are the close part and the side chains point away.
m_face_tm	geometric	field moment	Å	Mean Cα-Cα minus Cβ-Cβ distance over the contacting TCR:MHC residue pairs.

descriptor	class	operator	units	definition
ca_cb_agreement_tp	geometric	correlation	ratio	Spearman correlation between the Cα and Cβ distance maps over the TCR:peptide approach shell. High when the side chains track the backbone, as they do in a crystal.
ca_cb_agreement_tm†	geometric	correlation	ratio	The same rank correlation across the TCR:MHC approach shell.
fp_b0_r7	topological	homology invariant	count	Betti-0 of the flag complex on the contacted pMHC Cα atoms at 7 A: how many disconnected patches the footprint falls into.
fp_b1_r7	topological	homology invariant	count	Betti-1 at 7 A: how many holes the footprint encloses.
fp_chi_r7†	topological	homology invariant	count	Euler characteristic b0 - b1 at 7 A.
fp_b0_frac_r7	topological	homology invariant	fraction	Patches per contacted residue at 7 A; the size-free form of Betti-0.
fp_b0_r8	topological	homology invariant	count	Betti-0 of the flag complex on the contacted pMHC Cα atoms at 8 A: how many disconnected patches the footprint falls into.
fp_b1_r8	topological	homology invariant	count	Betti-1 at 8 A: how many holes the footprint encloses.
fp_chi_r8†	topological	homology invariant	count	Euler characteristic b0 - b1 at 8 A.
fp_b0_frac_r8	topological	homology invariant	fraction	Patches per contacted residue at 8 A; the size-free form of Betti-0.
sc_shape	geometric	correlation	ratio	Pearson r between the pMHC and TCR height fields over the shared grid; positive is complementary, the receptor riding up where the groove rises. Lawrence & Colman's Sc is the same idea on a dot surface.
sc_charge	compositional	correlation	ratio	Pearson r between the two charge fields; NEGATIVE is complementary, plus meeting minus.
sc_phobic	compositional	correlation	ratio	Pearson r between the two Kyte-Doolittle fields; positive is complementary, apolar meeting apolar.
sc_charge_prod	compositional	field moment	ratio	Mean per-cell product of the two charge fields.
sc_phobic_prod	compositional	field moment	ratio	Mean per-cell product of the two hydropathy fields.
sc_gap_mean	geometric	field moment	Å	Mean of h(TCR) - h(pMHC) over retained cells. Negative on a real interface: the median cell interdigitates.
sc_gap_sd	geometric	field moment	Å	Spread of the same gap. High when the receptor rests on a few high points rather than meshing.
sc_gap_vol	geometric	field moment	Å^3	Void volume, the gap integrated over the contact plane where it is positive.
sc_interlock	geometric	field moment	Å^3	Interdigitated volume, the gap integrated where it is negative. The larger of the two on a real interface.
sc_gap_index	geometric	field moment	Å	Void volume over retained contact area; the intensive form of the gap-volume channel.

descriptor	class	operator	units	definition
sc_interlock_frac	geometric	field moment	fraction	Share of retained cells whose gap is negative; the per-structure form of the corpus 71% interdigitation.
sc_gap_depth	geometric	field moment	Å	Mean depth over the interlocked cells alone: how far the receptor reaches in where it does.
sc_gap_height	geometric	field moment	Å	Mean standoff over the void cells alone: how high it stands where it does not mesh.
sc_gap_asym	geometric	field moment	signed fraction	(void - interlock) / (void + interlock); -1 for a face that only interlocks, +1 for one that only stands off.
sc_dh	geometric	field moment	Å	Mean absolute per-cell height difference between the two faces.
sc_dcharge	compositional	field moment	ratio	Mean absolute per-cell charge difference between the two faces.
sc_dphobic	compositional	field moment	ratio	Mean absolute per-cell hydropathy difference between the two faces.
sc_cells	geometric	count or share	count	Grid cells entering the comparison; bookkeeping, so a low complementarity can be told from a thin one.
sc_coverage	geometric	count or share	fraction	Retained cells as a share of the occupied pMHC cells in the window; bookkeeping.
co_pep	compositional	count or share	ratio	Contact order on the peptide: mean sequence separation of the peptide residues one CDR loop reaches, averaged over loops and divided by the peptide's span.
co_mhc	compositional	count or share	ratio	Contact order on the MHC helices, by the same construction.
partcoef_tcr	compositional	Hill number	fraction	Mean over engaged TCR residues of 1 - sum_s (k_s/k)^2 with the modules peptide and MHC; 0 when every residue reads one target only.
partcoef_pmhc	compositional	Hill number	fraction	The same over engaged pMHC residues with the six CDR loops as modules.

6.4 The energetics family (15 descriptors)

Sums of a pair potential over a contact set, and differences of them.

descriptor	class	operator	units	definition
Phi_tcr_pep	energetic	potential sum	log-odds	Φ over TCR-peptide contacts under TCRen2, summed over all TCR regions. Lower is more favourable.
Phi_tcr_mhc	energetic	potential sum	log-odds	Φ over TCR-MHC contacts under Miyazawa-Jernigan.
Phi_cdr12	energetic	potential sum	log-odds	The CDR1 + CDR2 part of the TCR:peptide energy, both chains.
Phi_cdr3a	energetic	potential sum	log-odds	The CDR3α part of the TCR:peptide energy.
Phi_cdr3b	energetic	potential sum	log-odds	The CDR3β part of the TCR:peptide energy.
dPhi_tcr_pep	energetic	potential sum	log-odds	Poly-alanine reference delta of the TCR:peptide energy; the pose-geometry baseline removed.

descriptor	class	operator	units	definition
dPhi_pep_soft	energetic	potential sum	log-odds	Smoothed reference delta, peptide direction: the peptide's energy minus the free energy of the residue background at each peptide position, receptor frozen.
varPhi_pep_soft	energetic	potential sum	log-odds^2	Variance of the local field under the background, peptide direction: how sharply each peptide position's energy responds to residue identity, summed over positions. NOT a ddG – it is a second cumulant, not a difference of differences.
dPhi_tcr_soft	energetic	potential sum	log-odds	Smoothed reference delta, receptor direction: the receptor's energy minus the free energy of the residue background at each contacted TCR position, peptide frozen.
varPhi_tcr_soft	energetic	potential sum	log-odds^2	Variance of the local field under the background, receptor direction, summed over contacted TCR positions.
dPhi_tra_soft	energetic	potential sum	log-odds	The α-chain part of the receptor-direction smoothed reference delta.
dPhi_trb_soft	energetic	potential sum	log-odds	The β-chain part of the receptor-direction smoothed reference delta.
Phi_pep_mhc†	energetic	potential sum	log-odds	Φ over peptide-MHC contacts under Miyazawa-Jernigan. Computed without the receptor.
dPhi_pep_mhc†	energetic	potential sum	log-odds	The same reference across peptide:MHC. Computed without the receptor.
Phi_pep_int†	energetic	potential sum	log-odds	The peptide's own intra-chain contact energy. Computed without the receptor.

6.5 The potts family (5 descriptors)

The same interface read against the partition function of the coupled contact model.

descriptor	class	operator	units	definition
neg_energy†	energetic	potential sum	kT	-E of the observed contact map under the coupled Potts model; higher is more native-like. Exactly log_z + log_lik.
log_z	energetic	potential sum	kT	Log partition function over every contact map the geometry admits; the interface's capacity.
log_lik	energetic	potential sum	kT	Log probability of the observed contact map; its typicality.
psi	energetic	potential sum	kT/site	log_lik per available site, so interfaces of different size compare.
n_contacts	energetic	count or share	count	Available residue pairs that engaged. Distinct from the footprint's loop tally.

6.6 The kinetics family (17 descriptors)

The contact map as a network of breakable springs. No potential enters.

descriptor	class	operator	units	definition
exp_lost	geometric	field moment	count	Expected TCR:peptide contacts lost under a 1 A isotropic shift.
mean_margin	geometric	field moment	Å	Mean contact margin, cutoff minus minimum heavy-atom distance.
frac_robust	compositional	count or share	fraction	Share of TCR:peptide contacts with at least 1 A of margin.
K_tens	geometric	spectral invariant	N/m	Tensile stiffness along the docking axis.
K_shear	geometric	spectral invariant	N/m	In-plane stiffness, S_tot minus K_tens.
aniso†	geometric	spectral invariant	ratio	K_shear / K_tens; how much stiffer the interface is along the pull than across it.
lam_max	geometric	spectral invariant	N/m	Largest eigenvalue of the stiffness tensor.
lam_min	geometric	spectral invariant	N/m	Smallest eigenvalue of the stiffness tensor.
n_spring	compositional	count or share	count	Springs in the interface network; every other kinetics column is NaN below three.
S_tot†	geometric	spectral invariant	N/m	Trace of the stiffness tensor; total interface stiffness.
rupture_force	geometric	mechanical work	N	Peak resisting force under steered separation along the weaker axis.
rupture_work	geometric	mechanical work	J	Force integrated to full separation; the off-rate proxy, and a geometry-only quantity no potential enters.
couple_pep	compositional	count or share	count	Peptide residues contacting both the MHC and the TCR.
couple_mhc	compositional	count or share	count	MHC residues contacting both the peptide and the TCR.
couple_tcr	compositional	count or share	count	TCR residues in the Vα-Vβ interface that also contact the pMHC.
couple_total†	compositional	count or share	count	Sum of the three coupling counts.
n_interface	compositional	count or share	count	Interface residue count; the size denominator for the coupling counts.

† carries a STATUS flag; see Section 7.

7 What is known against each descriptor

27 of the 164 carry a STATUS flag. The catalogue's own rule applies: a name absent from here has no known defect, and presence is not a reason to drop the column, only to know what it is. *Determined* means an exact identity over other shipped columns, so the column adds no information but remains convenient. *No receptor* means the value is computable without the TCR present, so it is constant across every structure of one epitope on one allele and a receptor-ranking model reading it reaches the cohort label without reading an interface. *Length-coupled* records a measured correlation with peptide or CDR3 length together with the share of variance that survives removing both.

***stalled* (1).**

descriptor	what is known
ct_tp_salt_bridge	only 3 distinct values over 1,707 modelled complexes: a salt bridge across the TCR:peptide interface is rare enough that the count is almost always 0. The TCR:MHC counterpart ct_tm_salt_bridge does move.

***suspicious* (26).**

descriptor	what is known
`pitch`	reads the generator's confidence rather than the interface: it is docking_angles' incident_angle, and it out-discriminates every clean docking angle for that reason. Never use it as a feature.
`fp_chi_r7`	determined: χ = b0 - b1 at the same radius.
`fp_chi_r8`	determined: χ = b0 - b1 at the same radius.
`D1_cell`	determined: D1 = 12 ** H_cell.
`J_cell`	determined: J = H_cell * ln 12 / ln S_cell.
`offset`	determined: offset = hypot(shift_u, shift_w).
`n_loop_contacts`	determined: n_pep_contacts + n_mhc_contacts.
`neg_energy`	determined: log_z + log_lik.
`S_tot`	determined by K_tens and K_shear.
`aniso`	determined by K_tens and K_shear.
`couple_total`	determined: couple_pep + couple_mhc + couple_tcr.
`crossing`	determined: abs(crossing_signed).
`Phi_pep_mhc`	no receptor: constant across every structure of one epitope on one allele, so a receptor-ranking model reading it reaches the cohort label without reading an interface.
`dPhi_pep_mhc`	no receptor; see Φ_pep_mhc.
`Phi_pep_int`	no receptor; see Φ_pep_mhc.
`n_pep_int`	no receptor; see Φ_pep_mhc.
`mhc_class_bin`	no receptor: it is the MHC class, I or II. It is also constant on any single-class cohort – both receptor benchmarks are class I – so it contributes nothing there and separates the classes everywhere else.
`m_erank_tp`	peptide-length coupled: -0.547 / -0.105, 58.3 per cent of variance beyond both lengths. It reads the class I bulge.
`m_erank_tm`	CDR3-length coupled: -0.186 / -0.437, 69.4 per cent beyond both. It reads how much loop there is to spread over the helices.
`m_gap_tp`	peptide-length coupled: -0.322 / +0.041, 90.8 per cent beyond both.
`ca_cb_agreement_tm`	coupled to both lengths: -0.349 / -0.334, 70.5 per cent beyond them – the most length-loaded column here.
`degree_evenness_tp`	peptide-length coupled: -0.450 / -0.006, 88.7 per cent beyond both. It reads the class I bulge.
`frac_well_coordinated_tp`	peptide-length coupled: -0.440 / -0.064, 83.4 per cent beyond both; see degree_evenness_tp.
`g_even_tcr`	peptide-length coupled: -0.368 / -0.061, 91.0 per cent beyond both.
`g_loop_even`	CDR3-length coupled: -0.186 / -0.350, 87.7 per cent beyond both.
`g_comp_frac`	CDR3-length coupled: -0.032 / +0.280, 92.7 per cent beyond both.

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